



LS-W

Tungsten-Halogen Light Source

User Manual

Version 1.2

 **Important safety note!** Before starting operating the W light source, please read this guide carefully for important safety and user information about your equipment and for the terms of the Warranty. This guide is intended to help you be more comfortable and productive while using this equipment. Failure to follow the instructions for the proper use and care for this equipment can increase the risk of serious personal injury and/or equipment damage.

Sarspec products are developed to be safe and effective. In order to avoid personal serious injuries and/or equipment damage, the user must:

- Never look directly at the optical exit, intense focused light. Also, some UV radiation is present. Radiation from direct or reflected UV light may burn your skin and eyes. Use adequate eye protection.
- Do not open the equipment or get in contact with the circuit board. This equipment works with high voltage, up to 600 V.
- Never touch the components or their connections. The printed circuit board is assembled with electrostatically sensitive components.
- Connect the equipment to the correct power source and use proper optical fibers.
- Do not spill liquids in the equipment or use it in damp or harsh environments.
- Never use the equipment near flammable items.
- Be aware about the danger of electrical shock if not used properly.

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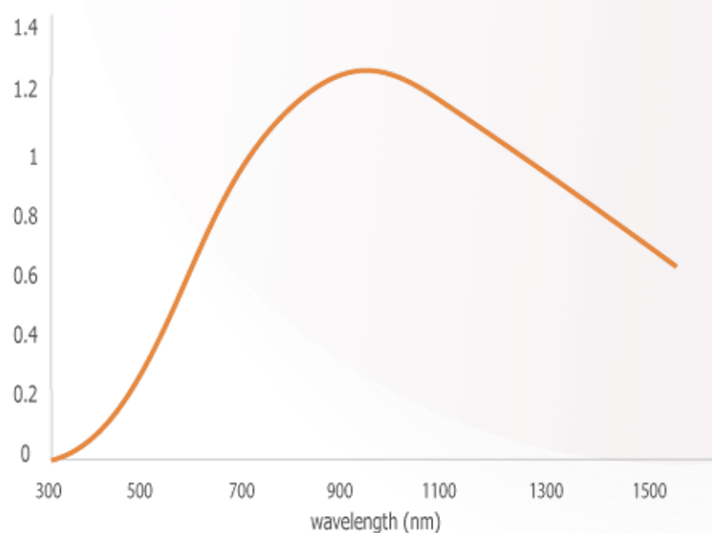




1 - LS-W with Filter Holder

About this product

LS-W is a compact, reliable and powerful tungsten light source that meets the demands for most spectroscopic applications covering the visible and near-IR spectral regions. The power output knob, the SMA connector with adjustable focus and the lamps with different powers (7, 14, 20, and 25 watts) provides the W light source with great flexibility and power efficiency that makes this light source the ideal solution for absorbance, transmittance, and reflectance measurements ranging from 300 to 2500 nm. The built-in shutter control simplifies the operation for dark measurements without the need to turn off the light source, while the built-in filter slot helps to accommodate filters for wavelength range selection.



2 – Spectral Irradiance of the Tungsten Lamp (LS-W20)

Main features

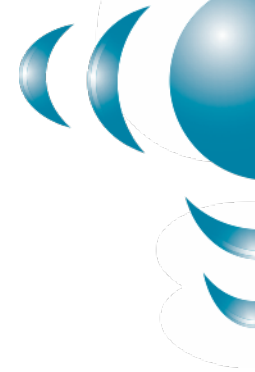
- Adjustable intensity knob.
- Highly stable output.
- Wide spectral range, from 300 to 2500 nm.
- Simple connection through optical fibers.
- Four lamps available (7, 14, 20 and 25 W).
- Local and remote operation.
- Built-in filter slot for slide filter (12.5 mm or 1/2").

Items Supplied

- W Light Source.
- AC Power Adapter.
- Slide Filter Support
- Product Report and User Manual (USB Pen Drive).

Optional Items

- Remote Cable.
- SPEC Hub.
- Bandpass, Shortpass, and Longpass Filters.
- Light Source Controller Software.



Operating instructions

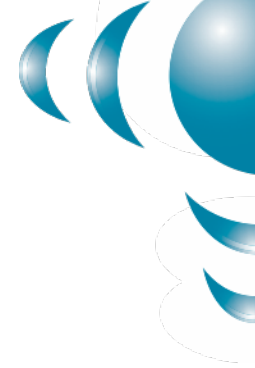
1. Plug the AC power adapter into a wall outlet
2. Plug the other end of the AC power adapter cable into the 15 VDC / 2 ADC socket on the rear panel of the W light source.
3. Plug the D-Sub 15-pin F connector of the remote cable (optional item) into the REMOTE connector on the rear panel. This point is only necessary if the light source is to be operated remotely.
4. Connect the optical fiber to the SMA 905 connector located on the front panel.
5. Select LOCAL or REMOTE operation of the light source with the switch located on the rear panel. Select LOCAL position for the operation with the switches on the rear panel, and REMOTE position for the operation with the remote cable (optional item). In the REMOTE position, the switches on the rear panel are disabled.
6. Set the lamp intensity by adjusting the INTENSITY knob located on the rear panel.
7. The SHUTTER switch located on the back panel blocks/unblocks the light exit. Select the ON position to block (close) the light exit and the OFF position to unblock (open) the light exit.
8. Insert the slide filter support into the slot located on the side panel
9. Switch on the power by turning the POWER switch on the rear panel (the Sarspec logo on the top of the light source will light up).



3 - Rear Panel (left) | Side/Front Panel (right)

Notes:

The Tungsten lamp ignites instantly following the switch-on.
Allow a 5 minutes warm-up time before starting measurements.

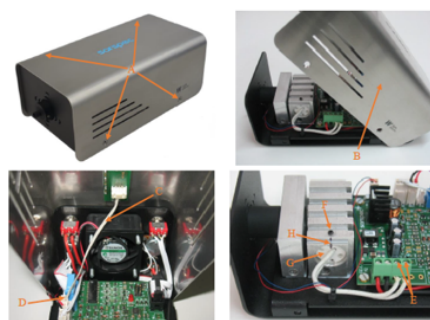


Tungsten lamp replacement

⚠ Important safety note! Before replacing the lamp unit, disconnect the W light source from your power source and allow the unit to cool for at least 20 minutes, if necessary!

The tungsten lamp unit must be replaced when it fails or finishes its service lifetime. We suggest contacting Sarspec for the replacement of the lamp unit. In the condition, if you want to replace it, please proceed as follows: (figure Erro! A origem da referência não foi encontrada.) :

1. Unscrew the four bolts (A) located on both sides of the equipment.
2. Carefully lift and rotate the cover (B) into the back direction. Be careful to avoid stressing the LED cable of the Sarspec logo (C).
3. Disconnect from the plug the LED cable of the Sarspec logo (D).
4. Remove completely the cover (B).
5. Unscrew the two bolts (E) and remove the tungsten lamp wires.
6. Pull out the lamp unit (G) and place it in a soft surface.
7. Carefully insert the new lamp until it stops ensuring that the black marks (H) are aligned
8. Tighten back the bolt (F). Be careful to not broke the ceramic socket of the lamp unit (G).
9. Insert the tungsten lamp wires (no polarity) and tighten the two bolts (E).
10. Connect the plug (D) of the LED cable of the Sarspec logo and close the cover (B). Be careful to avoid stressing the LED cable.
11. Screw the four bolts (A) of the cover located on both sides of the equipment



4 – Left View Side (top left) | Cover Opened into back direction (top right) | Inner view with connection of the LED of the Sarspec logo (bottom left) | Tungsten Lamp Unit and Mainboard (bottom right)

Note:

Please refer to the spare parts at the end of the manual for replacement lamp unit part number.

Maintenance

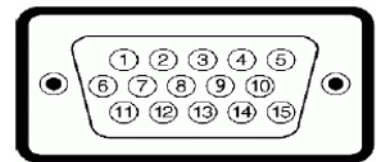
Warning! Unplug the Light Source before cleaning the equipment. Clean the external surfaces using a soft cloth. Do not use any abrasive cleaners. Do not allow any liquid to drip into the unit.

Remote Connector

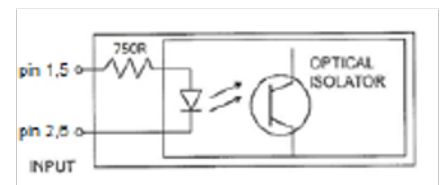
The remote connector is a 15-pin D-Sub Male type, with the pin assignment per table below.

1 - Pin Assignment

Pin	Description
1	Tungsten-Halogen lamp (input): 0V=off 5V=on (Optically isolated input)
2	Return for pin 1
5	Shutter (input): 0V=on (closed) 5V=off (open) (Optically isolated input)
6	Return for pin 5
7	Intensity control: 0 – 5 VDC or PWM (Useful range: 1 – 5 VDC)
8	GND (AUX. Output)
9	5V/500mA (AUX. Output)
3, 4, 10-15	NC (No Connection)



5 - Remote Connector: 15 pin D-Sub Male





Technical data / Specifications

Parameter		LS-W7	LS-W14	LS-W20	LS-W25
Applications		Absorbance, Transmittance, Reflectance and Irradiance			
Wavelength Range		380-2500 nm	380-2500 nm	300-2500 nm	380-2500 nm
Tungsten-Halogen Lamp	Power	7 W	14 W	20 W	25 W
	Lifetime	~ 4000 h	~ 4000 h	~ 1500 h	~ 4000 h
	Color Temperature	2800 K	2800 K	2950 K	2800 K
	Luminous Flux	105 lm	240 lm	350 lm	500 lm
	Bulb UV Filter	Yes	Yes	No	Yes
	Power Consumption (Max.)	12 W	19 W	25 W	30 W
Warm-Up Time		Approx. 15 min			
Operating Temperature		0 to 40 °C			
Lamp Cooler System		Air cooling			
Remote Connector		15-pin D-Sub Male (See remote connector table for pin assignment)			
Remote Operation		Shutter, Lamp Enabling (TTL compatible signals) and Intensity Control (Analog and PWM) – best with optional PEC Hub Useful Range: 1-5 VDC, Optical Intensity from 0.3% to 100%			
Local Operation		Shutter controller (using the local switch) Intensity Control (using the local knob, optical intensity from 15% to 100%) Remote/Local selector			
Optical Fiber Connector		SMA 905			
Humidity		95% RH, non-condensing			
Power Requirements		24 VDC/ 2 ADC			
Dimensions		200 (L) x 92 (W) x 72 (H) mm			
Weight		1.1 kg			



Troubleshooting

Problem	Possible Solution
The equipment does not turn on	• Ensure that the AC power adapter is plugged into the wall outlet and properly connected to the back panel. • Ensure the power switch on the back panel is on (the Sarspec logo on the top will illuminate).
No light output	• Ensure the optical fiber is correctly connected and not broken. • Ensure the shutter is OFF (open). • Ensure the operation switch is on the LOCAL position or if on the REMOTE position there is a 5V signal at pin 1 of the remote connector. • Check the cooling fan is rotating. The lamps are turned off if the air flow is not sufficient. • Replace the lamp unit.

Spare parts

Description	Part-Number
W Light Source 380-2500nm (Power Supply Included)	LS-W7
W14 Light Source (Power Supply Included)	LS-W14
W20 Light Source (Power Supply Included)	LS-W20
W25 Light Source (Power Supply Included)	LS-W25
Tungsten-Halogen Replacement Lamp (7W)	B-W7
Tungsten-Halogen Replacement Lamp (14W)	B-W14
Tungsten-Halogen Replacement Lamp (20W)	B-W20
Tungsten-Halogen Replacement Lamp (25W)	B-W25
AC Power Supply Adaptor	PS-W
SPEC Hub (cable included)	HUB-LS
Slide Filter Holder (for 12.7 mm diameter filters and 3mm max. thickness)	SFH-W



Other accessories available

Description	Part-Number
Spectrometer (FLEX STD UV/Vis/NIR)	FLEX-STD-UV-Vis-NIR
Optical Fibers	OF-UV-YYY-ZZZ* *YYY = Core Diameter ZZZ = Length
Reflection Probes	RP-NIR-YYY-ZZZ* *YYY = Core Diameter ZZZ = Length
Standard Cuvette Holder	SCH
Standard Probe Holder	SPH



Warranty

This equipment has a one-year warranty. Any complaint relating to defects must be submitted in writing without undue delay and shall contain a detailed description of the defect. Warranty may be voided by any of the following: Improper installation or application of instrumentation; use with improper powering; improper system design, disassembling and tampering, exposure to excessive heat, moisture, dirt, dust, corrosion, or any other condition deemed deleterious by Sarspec. Lamps are not covered by warranty. For any other information, please contact Sarspec specialists.

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